



TEAM NISHKARSH



CleanKop AI

An AI, IoT & GPS -Based Smart Waste
Collection and Monitoring System

HACKATHON PRESENTATION

PRESENTED BY

Parth Desai

Team Leader

Divya Pawar

Shreyash Patil

Tanavi Pawar

Official Problem Statement

"Waste collection operations often suffer from inefficient route planning, poor vehicle monitoring, and delayed grievance resolution, leading to inconsistent sanitation services. Develop a smart waste management platform that optimizes collection operations and improves service monitoring."

Current Challenges



Inefficient Collection Routes

Fixed routes instead of demand-based



Overflowing Garbage Bins

No real-time bin level monitoring



No Vehicle Tracking

No live GPS monitoring of trucks



Delayed Complaint Resolution

Manual handling, often ignored



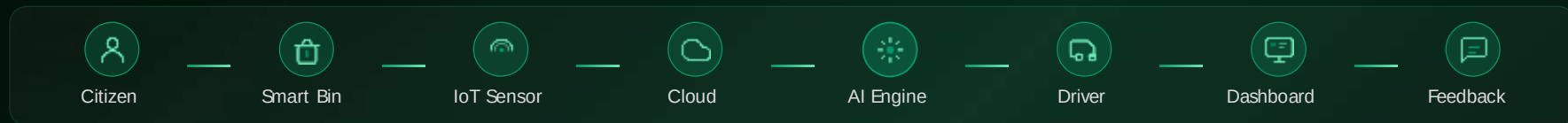
No Data Analytics

No performance dashboard

Current waste collection systems are reactive rather than proactive.

Introducing CleanKop AI

End-to-End Smart Ecosystem



Smart IoT Bins

Real-time fill level monitoring via ultrasonic sensors



AI Route Optimization

TSP + 2-Opt algorithm for shortest collection routes



GPS Vehicle Tracking

Live location monitoring for all collection vehicles



Citizen Complaint Portal

Photo + GPS complaints with auto-escalation at 48h



Real-Time Monitoring

Socket.IO powered live admin dashboard updates



Performance Analytics

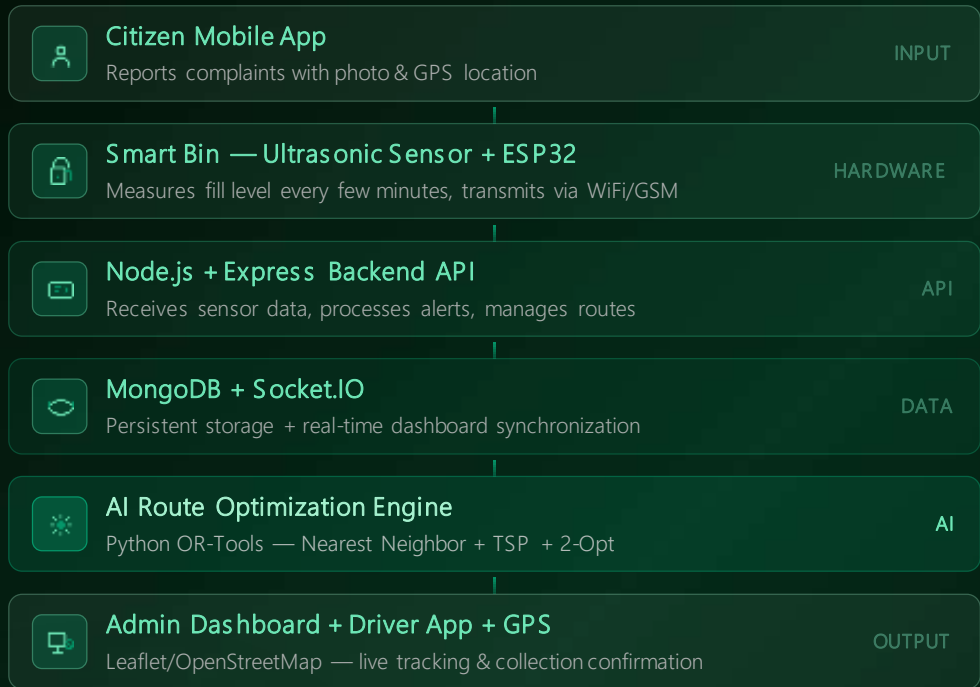
Heatmaps, fuel savings, collection efficiency reports

"CleanKop AI transforms traditional waste collection into an intelligent, automated, data-driven ecosystem."

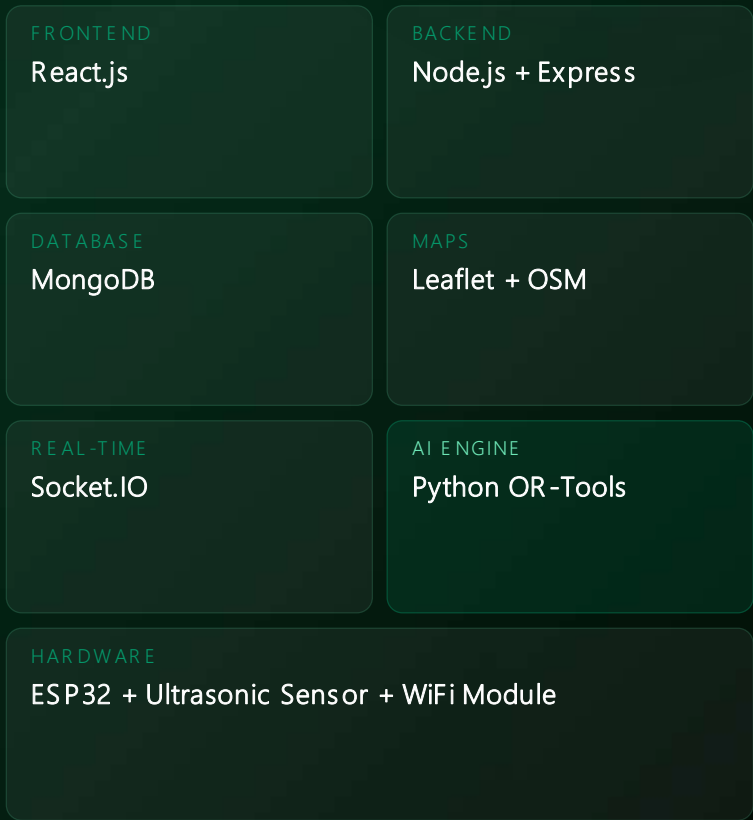
System Architecture

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Complete Data Flow



Technology Stack



How CleanKop AI Works

Complete 7-Phase Working Process

1 Smart Bin Monitoring

Citizen discards waste into smart bin

Ultrasonic sensor measures fill level 0–100%

ESP32 transmits data via WiFi/GSM to cloud

2 Backend Processing

Data stored in MongoDB, fill % calculated

Auto alert generated when fill exceeds 80%

Socket.IO instantly updates Admin Dashboard

3 AI Route Optimization

Bins prioritized: Red >80%, Yellow 50–80%

TSP + 2-Opt generates shortest route

Route assigned to nearest available vehicle

4 Driver Operations

Driver receives optimized route on mobile app

Live GPS tracking throughout collection

Bin marked collected; status resets to 0%

5 Citizen Complaint System

Citizen reports overflow with photo + GPS

Auto ticket ID generated; admin assigns driver

Auto-escalation if unresolved beyond 48 hours

6 Monitoring & Analytics

Daily collections, complaint stats, fuel savings

Heatmaps, truck utilization, driver history

Collection efficiency and performance metrics

7 Smart Alerts

- Bin Full Alert — fill above 80%
- Complaint Escalation — beyond 48 hours
- Vehicle Idle Alert — stationary 30+ minutes

| Key Features & Benefits

| Key Features

- ✓ AI Route Optimization
- ✓ Real-Time GPS Tracking
- ✓ IoT Smart Bins
- ✓ Citizen Complaint Portal
- ✓ Live Admin Dashboard
- ✓ Heatmaps & Reports
- ✓ Performance Analytics
- ✓ Automated Smart Alerts

| Expected Impact

- 30% Reduction in Fuel Consumption
- 80%+ Live Bin Monitoring Coverage
- <48h Complaint Resolution Time
- 100% Fleet Visibility via GPS
- 3x Faster Waste Collection Cycles
- 40% Lower Operational Cost
- 95% Citizen Satisfaction Score

"Turning traditional waste collection into a smart, connected, and sustainable urban ecosystem."



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THANK YOU

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"Smart Technology for a Cleaner Tomorrow"